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EXPLAIN COMOUTER ARCHITECTURE CONCEPTS

US 14917

FORMATIVE ASSESSMENT

QUESTION 1

Computer architecture is a description of the design of a computer system. It has a physical and abstract level, in short it refers to how a computer system is designed and what technologies it is compatible with.

Computer architecture is the design and organisation of the fundamental structure and operational principles of a computer system. It is a blueprint that describes how hardware and software interact, how components are connected, and how data is processed. It is often divided into 3 categories : ISA , Micro-architecture and system architecture

Elements

1. Instruction set architecture (ISA)

   Function: defines the set of instructions the processor can understand, data types, registers, addressing modes

   Categories: CISC Complex instruction set computer and RISC Reduced instruction set computer, example are our phones use ISA because it needs low power consumption

2. Microarchitecture

   Function : how the ISA is physically implemented inside the processor, defines data path, data processing and storage elements

   features: pipeline, execution units, cache hierarchy and clock speed, intel core i7 is an example

3. **System architecture**

   Function : defines how major components interconnect (CPU, Memory)

   Categories: Von Neumann architecture has a single memory for data and instructions, Havard separate memory

   Feature : bus structure , memory hierarchy, example is a bank ATM, the same RAM holds both the program to count cash and the account balance data

QUESTION 2

1. INPUT UNIT: keyboard gets data in
2. CPU CENTRAL PROCESSING UNIT acts as the brain , it executes instructions, performs calculations and controls other parts

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3. Output unit consists of devices with the help of which we get the information from computer such as a monitor
4. Random access memory holds data of currently running programs , for example multiple tabs in chrome open
5. Motherboard is the main circuit board that connects and allows communication between all components

Question 3

The **MOTHERBOARD** acts as the central highway system and the city that provides physical sockets for CPU, RAM and connectors for buses

The **CENTRAL PROCESSING UINIT** acts as the traffic controller and factory , it initiates operations, fetch instructions from RAM , decodes them an executes them

Buses are actual roads that carry data

Address bus CPU says where to get data from

Data bus carries the actual data

Control bus carries how control signals like read, write, clock

Flow example: you click an app icon

1. Motherboard's bus connect solid state drive to RAM to CPU
2. CPU puts address of an app on Address Bus via motherboard
3. Control Bus sends a "Read" signal
4. 4 App data travels from storage via Data bus into RAM ,then again via data bus into CPU registers
5. CPU processes it and sends result via data bus through motherboard to GBU/monitor

Without motherboard there's no connection, Without Central Processing unit there's no control, without buses there's no transport

QUESTION 4

1. Instruction length
2. Memory transfer
3. Word length
4. Instruction format
5. Instruction fetch
6. Operand Specification

How they affect how computers process information

1. Instruction length constraint: if Instruction length is fixed, decoding is very fast and pipelining is easy ,so CPU processes faster but it limits how much information you can put in one Instruction

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2. Memory transfer: RAM is used by computers to process data and transfer it into correct areas of Memory so the transfer 9f data into and out of the Memory is an important part of the design of a computer's Instruction set architecture
3. Word length refers to the number of bits that a computer groups together, when it transfers or moves the data it is processing while carrying out an Instruction, the larger the groups the more complicated the Instruction will be

QUESTION 5

1. Use microprogramming whereby complex instructions are broken into simple micro-operations to accommodate hardware instructions
2. Optimize compiler design: compiler can be made intelligent to use registers efficiently, reorder instructions and choose best addressing modes to overcome register and Memory constraints
3. Pipelining and parallelism: to overcome performance constraints, break instruction execution into stages so multiple instructions are processed at once
4. Layered and modular design : Keep ISA simple and move complexity to software which makes it easier to accommodate power and cost constraints
5. Use caching and virtual memory: to accommodate limited fast memory, use cache hierarchy and virtual memory to give illusion of large fast memory
6. Maintain backward compatibility: introduce new instructions as extensions without removing old ones so old software still works while accommodating new needs